One plan you can trust,
then an agent that helps you change it.

Ferronova · portfolio, scarce resources and re-planning on the Databricks lakehouse

Rui Dias · Senior FDE take-home

THE PROBLEM AS WE UNDERSTAND IT

Thin margins, hard penalties, shared scarce resources, planned in spreadsheets

Margin is fragile

- Single-digit operating margins
- Penalties capped around 5 to 10% of contract
- One slipped milestone can erase a programme's profit

Risk spreads silently

- A few rigs and crews serve 20 to 25 programmes
- One rig down moves risk to other programmes
- Nobody sees it until a date is missed

Planning lives in heads

- Excel and local factory tools
- Constraint knowledge in a few senior planners
- Key-person risk on the core of the business

The question nobody can answer in less than days today

“If this bid lands, or this test rig goes down, what happens to everything else, and to our margin?”

WHAT CHANGES

From reconciling spreadsheets to deciding on one live picture

	Today	With the system
Portfolio view	Assembled by hand, weeks old	Live: every programme, milestone and scarce resource
Rig and crew bookings	Trackers per rig and per crew	One plan of record. A double booking cannot be saved
“What if this bid lands?”	Days of planner work	A scenario in minutes, with penalty exposure
Rig goes down	Phone calls, re-plan over days	Agent proposes a re-plan the same day, planner decides
Who decided what	In email threads	Every commit versioned: who, why, at what cost
Planner expertise	In heads	Captured as rules and data, still under planner control

BUSINESS VALUE

Protect margin first, then grow the order book safely

Lever	How	Order of magnitude / year
Avoid penalties	Risk visible weeks earlier, same-day re-plans	EUR 5 to 10M
Recover from disruptions	~2 weeks recovered per major disruption	EUR 2 to 4M (partly overlaps)
Bid on dates we can meet	Bids checked against real rig and crew capacity	Avoid EUR 70M tail risk on one large bid
Use scarce assets better	Fewer idle rig gaps, crew moves, parked WIP	EUR 0.5 to 2M + cash
Planner capacity and resilience	Scenarios in minutes; knowledge captured	Capacity, lower key-person risk

How to read it

- Assumes EUR 7.7bn in flight, 6% margin, penalties ~1% of revenue
- Penalty lever alone ≈ 3 to 7% of operating profit
- Board-level case: bid risk and resilience
- Build ≈ EUR 1 to 1.5M, payback < 1 year if penalty assumption holds

First number to get in week 1: penalties actually paid or accrued in the last 3 years, by milestone. If it is 0.3% of revenue, not 1%, the case rests on bid risk and resilience.

THE ORDER WE UNLOCK IT

Each step pays for itself and makes the next one safe. The agent comes last.

1 · One picture

Weeks 0–8

- Sources in, IDs resolved
- Portfolio risk dashboard
- Read-only, low risk

2 · Plan of record

Weeks 6–16

- Rig, crew, hall bookings
- Hard rules enforced
- Trackers retired

3 · What-if

Weeks 12–20

- Bid and disruption scenarios
- Penalty exposure per option
- Better bid decisions

4 · Agent proposes

Weeks 18–26

- Rig-down re-plan first
- Planner approves or edits
- Days to hours

Where we push back

The ask is an agent on top. On today's data it would reason over spreadsheets that disagree. Steps 1 and 2 deliver most of the risk reduction and make the agent safe. Later: margin and cash forecast, portfolio optimisation, factory line balancing.

HOW IT STAYS SAFE

The agent proposes. The data model refuses what is impossible. A person decides.

Rig A breaks down

10-week outage recorded. Two test campaigns now conflict.

Agent proposes

Moves both campaigns to other capable rigs.
Cost: type approval 59 days late, EUR 2.95M.

Planner improves it

Pulls another campaign forward.
Cost down to EUR 1.4M, 28 days late.

Director approves

Above the penalty threshold, so a director must sign.

Committed as plan v2

One transaction, versioned, conflicts cleared.

What the system refused along the way (all tested)

Attempt	Refused because
Book two campaigns on the same rig at the same time	A rig hosts one campaign at a time
Send a crew not qualified for the vehicle	Crew qualification checked on the dates
Test outside the customer's night or weekend window	Must sit inside a granted window
Agent changes a contract date or commits its own plan	No permission. Only planners commit

WHAT MAKES OR BREAKS IT

Three things decide success. Only one of them is technical.

1 · Planner adoption

- Planners see their own rules in the model
- They keep the final say, always
- Co-design from week 1
- Measure: % of changes made through the system

2 · One write path

- Every rig, crew and hall change goes through the plan of record
- Trackers retired, not kept in parallel
- A management decision, not a technical one

3 · Data you can trust

- Every factory names things differently
- ID crosswalk with named owners per source
- Unresolved records visible on the dashboard, never silently dropped

OPEN QUESTIONS

What I need from you to firm this up

Business

- What did penalties cost in the last 3 years, by milestone?
- Which constraint binds most today: rigs, crews or hall space?
- How are penalties written: caps, recovery, force majeure for rig failure?
- Who may accept a plan that makes a customer milestone late?
- At what bid probability should a bid reserve capacity?
- How much autonomy for the agent in year one?

Technical

- Direct DB access with change capture on the scheduling tool?
- Which MES per factory, and what can each export, how often?
- Does the register cover batches, rigs and crews, or only programmes?
- Lakebase in your region, with the extensions we need?
- Planner and factory groups already in Entra ID?
- How fresh must the picture be: minutes, hours, daily?

HOW I WORKED WITH AI

Faster on the mechanical parts, overruled on the judgment calls

Where it moved me faster

- Drafted the Postgres DDL, seed data and a test harness in minutes
- I spent the saved time on the rules: which are hard, which are priced
- 15 executable tests prove the rules hold for agent and planner alike

Where I overrode it

- First draft enforced rules with Delta CHECK constraints and pipeline expectations. Those detect, they do not prevent cross-row conflicts
- First draft put agent safety in the prompt. Moved it into grants, row-level security and a single commit function

Adapt this slide to your own session before sending.

Backup · architecture, data model, agent, planner view

BACKUP · ARCHITECTURE

Lakehouse for the picture, Lakebase for the plan of record

Sources	Ingest + conform	Plan of record	Read models	Surfaces + agent
• Scheduling DB • MES × factories • PLM • CRM • ERP + sheets • Planner trackers • Programme register	• Lakeflow Connect, Auto Loader • bronze: raw, lineage • silver: conformed on register IDs • ID crosswalk + quarantine • Expectations, SCD2	• Lakebase (Postgres) • Bookings: rig, crew, hall • Scenarios, approvals, versions • EXCLUDE, FK, CHECK, RLS • commit_scenario: gates G1–G6	• gold: milestone risk • resource load by week • scenario impact, margin • Metric views	• Databricks App: planner • AI/BI + Genie: leadership • Agent: service principal • Tools = UC functions • Optimiser = black box

- Why Lakebase: Delta enforces only row-level rules. “One campaign per rig at a time” is a cross-row rule. Postgres makes it impossible to save.
- Governance: Unity Catalog grants, row filters (factory), column masks (margin). Same identities in Lakebase. Agent has no write grant outside its draft.

Every hard rule is a database constraint, scoped by scenario

Rule	Mechanism
Rig: one campaign at a time	EXCLUDE (scenario, rig, period &&)
Rig can run the test	FK (rig, test type) → rig_capability
Crew qualified on those dates	FK (crew, design, qual_period) + CHECK period <@ qual_period
Inside customer access window	FK (window, site, window_period) + CHECK period <@ window
Crew in one place, travel included	EXCLUDE (scenario, crew, [mobilise_from, end) &&)
Hall never over-committed	Bays as positions + EXCLUDE (scenario, position, period &&)
No commit on broken rig / revoked window	Commit gate G4 over v_plan_conflicts
Milestones and penalties	Priced per scenario; G5 escalates above threshold
No stale overwrite	base_version check under lock on plan_head

Core tables

- Reference: programme, design, design_type_test, milestone, campaign
- Resources: test_rig, rig_capability, crew, crew_qualification, access_window, hall_position
- Plan (by scenario): rig_booking, crew_assignment, hall_occupancy
- Governance: scenario, approval, plan_version, operational_event, guardrail_policy

Scoped to one trigger first: a test rig goes down

Proposes

- Re-sequence homologation campaigns across capable rigs
- Next: place a landed bid's campaigns and crews
- Reads via UC functions: affected bookings, capable rigs, free slots, impact
- Writes only into its own draft scenario

Never touches

- Contract dates and penalty terms
- Customer windows, qualifications, rig capabilities
- Factory shop-floor sequencing
- The committed plan, approvals, thresholds
- Bid pricing and customer talks

Guardrails live in the database

- RLS: own draft only
- Constraints: every row feasible
- No EXECUTE on commit
- G2 stale, G4 reality, G5 ruinous → director
- MLflow trace + rationale + row diff for audit

BACKUP · PLANNER VIEW (MOCK)

How a proposed re-plan appears, and where the planner decides

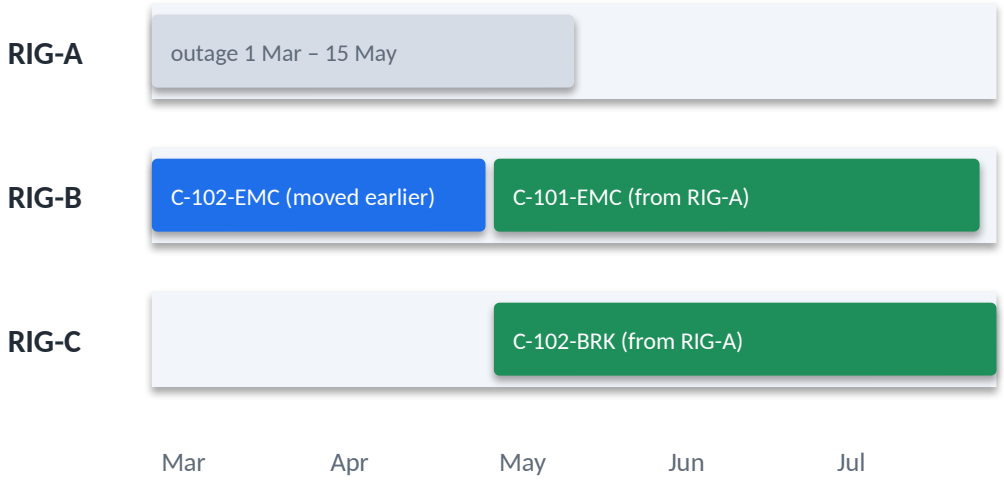
Planner workbench · Plan v1 · 1 open event · 1 proposal awaiting review

RIG-A down · 1 Mar – 15 May
2 campaigns in conflict
Reported by TC-1 maintenance log

Impact vs plan v1

Penalty delta: + EUR 1.40M
Newly late: M-101 Type approval (+28 d)
Unscheduled campaigns: 0
RIG-B load May–Jul: 100%
Needs: programme director

Proposed bookings (diff vs plan v1)



Agent rationale: only EMC-capable rig free is RIG-B; BRAKE moves to RIG-C same dates.
Edited by planner_ana: C-102-EMC pulled to 1 Mar (−EUR 1.55M).

Approve & commit

Edit (drag bookings)

Request director sign-off

Reject with reason

Invalid drags are refused on the spot, with the rule that blocked them.